

Dmitry Messerman

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Professional Summary

Electrical Engineer (Ph.D.) with 30+ years of experience in **complex system modeling, multidisciplinary engineering, and large-scale solution development**.

Proven expertise in **electrical system analysis, power transmission modeling, and physics-based simulation**, combined with extensive hands-on leadership of **end-to-end system engineering activities**, from requirement definition to deployment and validation.

Early career focused on **high-voltage power transmission systems**, including transient behavior, wave propagation, electromagnetic interference and environmental impact of electrical networks.

Later expanded into leading **complex multidisciplinary system development**, integrating electrical, thermal, and computational models for large-scale engineering platforms.

Seeking to apply deep system engineering and modeling expertise to **distributed energy systems (DER), smart grids, and advanced energy management solutions**.

Core Competencies

- Electrical System Modeling & Simulation (Power & Physical Systems)
- Transmission Line Theory & Transient Analysis
- Multidisciplinary Systems Engineering (Electrical, Thermal, Computational)
- System Requirements Definition & Specification
- Technical Leadership & Cross-Functional Coordination
- Optimization Algorithms (including Genetic Algorithms)
- Numerical Methods (PDE and BIE Solvers, FDM and BEM)
- System Validation, Debugging & Acceptance Processes
- Customer Requirement Analysis & Technical Translation

Professional Experience

Principal Engineer | Intel Corporation, Haifa

1992 – 2025

Led development and deployment of **complex engineering systems and modeling tools**, serving as technical authority across multiple multidisciplinary domains.

System Engineering & Technical Leadership

- Translated stakeholder and system requirements into **detailed technical specifications and working solutions**

- Led cross-functional engineering efforts across hardware, software, and modeling domains
- Defined system architecture and validation strategies for production-grade tools
- Coordinated development, deployment, and long-term support across multiple engineering teams

Modeling & Simulation (Electrical and Physical Systems)

- Developed **physics-based models of electrical and thermal behavior**, including:
 - Surrogate models for silicon thermal dissipation based on diffusion PDE solvers
 - Modeling of parasitic electrical effects in large scale high-speed systems based on $O(N)$ complexity algorithms
- Applied **numerical methods and system-level simulation techniques** for large-scale engineering problems
- Deployed novice multi-grid algorithm for CPU power grid (about 10^8 elements) simulation with $O(N)$ complexity

Optimization & Advanced Algorithms

- Led development of **genetic algorithm-based optimization methods** for improving system coverage and performance
- Led development of tools and algorithms for **parasitic resistance and capacitance networks synthesis from silicon design layout, reliability validation and variability analysis**

Multidisciplinary Engineering

- Integrated electrical, thermal, and computational models into **cohesive system-level solutions**
- Led HW/SW co-simulation platforms, enabling validation of complex system interactions

System Validation & Deployment

- Defined and executed **validation, debugging, and acceptance processes** for production systems
- Delivered tools used in production for up to 10–20 years

Software Engineer / Physicist | FrantzTech Ltd.

1990 – 1992

- Co-developed **signal processing algorithms** for ultrasound-based medical systems
- Applied advanced modeling techniques to improve detection of diffuse diseases such as NAFLD (Non Alcoholic Fatty Liver Disease)

Researcher | Institute for High-Voltage Direct Current Power Transmission

1982 – 1990

Worked on **large-scale electrical power systems**, focusing on transmission infrastructure.

- Developed algorithms for **modal analysis of multi-conductor transmission lines**, including real soil effects
- Analyzed **transient phenomena**, including lightning surge propagation and attenuation
- Studied environmental impact of **electric fields in high-voltage systems**
- Built mathematical models of **electromagnetic wave behavior across wide frequency ranges**

Education

Ph.D. in Electrical Engineering

St. Petersburg Polytechnic University

- Thesis: *Mathematical Modeling of Wave Processes in High-Voltage Power Transmission Lines across a wide range of frequencies and soil electrical parameters*

M.Sc. in Electrophysics

St. Petersburg Polytechnic University

Technical Skills

- **Programming:** Python, C/C++, MATLAB, Perl, Tcl
- **Modeling & Simulation:** Numerical methods, PDE solvers, system-level modeling
- **Tools:** Git, Linux environments
- **Languages:** English, Hebrew, Russian

Selected Publications & Patents

- Equalization Optimization Algorithm for Signaling Analysis (IEEE NEMO)
- Variability Analysis for System Robustness (SPIE)
- Inductance Modeling (US Patent)
- Reorganizing rectangular layout structures for improved extraction (US Patent)